Yulia Volkova

yulia-volkova.github.io

LinkedIn • GitHub • yuulia.volkova@gmail.com • +447450514095

I am a MATS 9.1 scholar working on AI control evaluations with Rhys Ward and Redwood Research. I have 7+ years of experience building insider threat detection and communication surveillance systems at Behavox for Tier-1 financial institutions. I first hand witnessed AI integration into the regulation sector, which inspired me to move and work on AI safety. Further details can be found on my website.

Education

London School of Economics
MSc Economics
- Distinction (dissertation in behavioural economics)
- Courses included Advanced Macroeconomics, Econometrics, Quantitative Methods

London School of Economics
BSc Philosophy & Economics
- Distinction
- Courses included Mathematical Methods, Statistics, Econometrics, Philosophy of Science

Selected Publications

2026 **Control Evaluations for Covert Malicious Fine-Tuning.**
Buckworth, T., Moreno, R., Panda, A., **Volkova, Y.** Supervised by Dr Francis Rhys Ward. *Submitted to NeurIPS 2026*

Residual Stream Decoders for Probing Model Activations.
Algoverse AI Safety Research Fellowship. PI: Callum McDougall, mentor: Nicky P. *Submitted to ICML*

Positions and Programs

MATS 9.1 Extension *Apr 2026 – Present*
- Continuing control evaluations research under Francis Rhys Ward

MATS 9.0 *Jan – Apr 2026*
- Control evaluations for covert malicious fine-tuning under Francis Rhys Ward, advised by Tyler Tracy and James Lucassen (Redwood Research)
- Selected for MATS 9.0 Spotlight Talk — opened the London event

Athena Women in AI Safety Research Fellowship — *Nov 2025 – Jan 2026*

- Mechanistic interpretability approach to monitoring chain-of-thought faithfulness under Arun Jose
- Gathered feedback from Adam Karvonen, Goodfire team, and Uzay Macar; integrating in MATS 9.1 extension phase

SPAR Fellowship — *Oct – Dec 2025*

- CoT monitorability project under Qiyao Wei (MATS 8.0, Cambridge University)

Visiting Member, London Initiative for Safe AI (LISA) — *Sept 2025 – Present*

Algoverse AI Safety Research Fellowship — *Jul – Aug 2025*

- Mechanistic interpretability under Nicky P
- Cooperative AI under Max Kleiman-Weiner

ARBOx2 Oxford AI Safety Bootcamp — *Jul 2025*

- Completed the ARENA curriculum

Media & Talks

• **MATS 9.0 Spotlight Talk** — opened the London event (YouTube)

• **MATS Project Talks at LISA and Google DeepMind London** (Jun 2026) — presented the covert malicious fine-tuning project (F. R. Ward's stream) to research audiences at both offices

• **FellowCast Podcast at Constellation** — "Covert Malicious Fine-Tuning" (Spotify)

Work Experience

Behavox — *2017 – 2025*

Insider threat detection, communication surveillance, and compliance

Data Science Production Team Lead — *2024 – 2025*

- Led development and deployment of LLM-driven Java applications across 3 products and 14 languages, using a pipeline of fine-tuned models (SBERT → DeBERTa → Qwen-14B)
- Managed execution of Java applications on the Apache Beam Dataflow runner; monitored GPU utilization and used BigQuery datasets to debug pipeline behavior and resolve incidents

- Led cross-team roadmapping for ML features under production constraints
- Built reusable tooling—CI/CD templates and MLflow tracking—enabling rapid MVPs for client deliveries

Insider Threat Product Lead *2020 – 2024*

- Designed risk taxonomy and classifiers for insider threats—data exfiltration, disgruntlement, flight risk
- Worked directly with heads of insider risk at Citadel, SoftBank, and other Tier-1 financial institutions to validate detection models against real threat landscapes
- Led R&D and production of high-recall (81% at 0.5% FPR) misconduct detection systems using RoBERTa + SBERT pipeline
- Validated that shorter text sequences in RoBERTa training maintain performance, doubling training speed and halving costs
- Built internal tooling (chatbot, recommendation system) serving 200+ daily users

Data Scientist *2017 – 2020*

- Landed proof of concepts that resulted in 3+ years client contracts
- Improved compliance model recalls from 70% to 90%
- Reduced client bug requests from 1/day to 1/month by setting up an efficient QA process
- One of the first Data Science hires; created foundational documentation during company scaling

Additional Skills and Interests

Programming Languages: Python, PyTorch, HuggingFace, Pandas, NumPy, Scikit-learn, FastAPI, spaCy, SQL, Java, Kotlin

Machine Learning & NLP: LLM fine-tuning (LoRA/qLoRA), prompt engineering, transformers, SBERT, PCA, t-SNE, Weights & Biases

Infrastructure: AWS S3, GCP, Docker, Nomad, Kubernetes, TeamCity, Gitlab CI/CD pipelines

Statistics & Econometrics: Hypothesis testing, time-series analysis, A/B testing, IV estimation, panel regression

Leadership: Mentorship, technical interviewing, product roadmapping, cross-team communication

Other Interests: Oil painting, skiing, hiking